Backward Chaining

Aim

To implement and demonstrate **Backward Chaining in Prolog** using medical diagnosis, student eligibility checking, and travel planning.

Exercise 1: Medical Diagnosis

Program

```
None
% Facts
has_fever(john).
has_cough(john).

% Rule
has_flu(X) :-
    has_fever(X),
    has_cough(X).
```

Query

```
None
?- has_flu(john).
```

Output

```
None
true.
```

Exercise 2: Student Eligibility Checker

Program

```
None
% Facts
passed(alice, maths).
passed(alice, science).

% Rule
eligible(X) :-
    passed(X, maths),
    passed(X, science).
```

Query

```
None
?- eligible(alice).
```

Output

```
None
true.
```

Exercise 3: Travel Planner

Program

```
None
% Facts
has_ticket(james).
has_visa(james).

% Rule
can_travel(X) :-
    has_ticket(X),
```

```
    has_visa(X).
```

Query

None
```
?- can_travel(james).
```

Output

None
```
true.
```

Result

Thus, **Backward Chaining** was successfully implemented in Prolog. The system started with the query (goal) and worked backward to check whether the required facts were available. All three queries were successfully proved.